American Computer Science League

2021-2022 ● Contest 1: Short Problems ● Senior Division

1. Computer Number Systems

Starting with the year 2122_{10} converted to hex, in how many years (in base10) will the year first be represented by a hexadecimal number where all the digits are the same?

A. 22
B. 62
C. 260
D. 406
E. 608

2. Computer Number Systems

The Fibonacci sequence {0, 1, 1, 2, 3, 5, 8, 13, …} is formed by adding the two previous terms: 5 + 8 = 13. Which number in the sequence when converted to hexadecimal would have the 2nd occurrence of the substring "0"?

A. 34
B. 55
C. 89
D. 144
E. 233

3. Recursive Functions

Find $f(10)$, given

$$f(x) = \begin{cases} 1 & if\ x = 1 \\ f(x-1) + 3x - 2 & if\ x > 1 \end{cases}$$

A. 70
B. 92
C. 117
D. 145
E. 176

4. Recursive Number Systems

Evaluate $f(100, 36) - f(36, 100)$, given

$$f(x.y) = \begin{cases} f\left(\left[\frac{x}{2}\right], \left[\frac{y}{2}\right]\right) + 2 & if\ xy \geq 50 \\ f(2x, y-3) + 1 & if\ 10 < xy < 50 \\ xy - x - y & if\ xy \leq 10 \end{cases}$$

where [z] returns the greatest integer less than or equal to z

A. 128
B. -134
C. -1
D. -125
E. -143

5. What Does This Program Do? (Branching)

What is the final value of x after this program is executed?

```
a = 1: b = 2: c = 3: x = 0
if a < b then
   x = x + a
else
   x = x + b
end if
if b > c && a <= b then
   x = x + c
else
   x = x + a
end if
if !(a < b && a < c) then
   x = x + a
end if
if b < c || c < a then
   x = x + c
end if
output x
```

A. 4
B. 5
C. 6
D. 7
E. 8

美国计算机科学联赛

2021-2022 ● 第一轮: 短题目 ● 高级组

1. 计算机数字系统

从转化为十六进制的 2122_{10} 年开始，多少年（十进制）后，十六进制后的年份将由相同的数字组成？

A. 22
B. 62
C. 260
D. 406
E. 608

2. 计算机数字系统

斐波那契数列 {0, 1, 1, 2, 3, 5, 8, 13, ...} 由前两项相加而成，例如：5 + 8 = 13。将数列中的每个数字转换为十六进制后，第二次出现子串 "0" 的是哪个数？

A. 34
B. 55
C. 89
D. 144
E. 233

3. 递归函数

根据以下条件， 求 $f(10)$。

$$f(x) = \begin{cases} 1 & x = 1 \\ f(x-1) + 3x - 2 & x > 1 \end{cases}$$

A. 70
B. 92
C. 117
D. 145
E. 176

4. 递归函数

根据以下条件，计算 $f(100, 36) - f(36, 100)$ 。

$$f(x.y) = \begin{cases} f\left(\left[\frac{x}{2}\right], \left[\frac{y}{2}\right]\right) + 2 & xy >= 50 \\ f(2x, y-3) + 1 & 10 < xy < 50 \\ xy - x - y & xy <= 10 \end{cases}$$

其中 [z] 返回小于或等于 z 的最大整数。

A. 128
B. -134
C. -1
D. -125
E. -143

5. 此程序是怎么运行的？（分支）

该程序执行后，x 的最终值是什么？

```
a = 1: b = 2: c = 3: x = 0
if a < b then
   x = x + a
else
   x = x + b
end if
if b > c && a <= b then
   x = x + c
else
   x = x + a
end if
if !(a < b && a < c) then
   x = x + a
end if
if b < c || c < a then
   x = x + c
end if
output x
```

A. 4
B. 5
C. 6
D. 7
E. 8